Venkata Lakshmi Parimala Pasupuleti

Software Engineer · AI Deployment Infrastructure · Python · Cloud · CI/CD · Distributed Systems · Observability · Linux · 9yr Production

+1-703-261-5410 | lakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Software engineer with 9+ years of production experience building cloud delivery infrastructure, CI/CD pipelines, distributed systems, and observability tooling - the infrastructure layer that tightens the feedback loop between development and real-world deployment. At Entrata, I owned production systems on AWS (S3, EC2, RDS, Glue) at 99.5% uptime across a multi-tenant SaaS platform: event-driven async architecture (RabbitMQ) with retry logic and idempotency guarantees, Docker microservices delivering 50% scalability improvement, Python and Bash automation pipelines eliminating 80% of manual operational workload, and GitHub Actions CI/CD maintaining deployment quality across 9 years of production operation. AgentProbe is the metrics and observability proof point: a production AI evaluation platform with a 5-metric instrumentation engine (correctness, latency, token efficiency, hallucination rate, groundedness), CI-integrated regression gate that blocks quality degradation before deployment, and multi-provider model routing - exactly the kind of metrics pipeline, dashboards, and deployment gate the AI Deployments team builds to maintain visibility from model to production. I build reliable systems that ship fast, instrument thoroughly, and stay up - on Linux, on AWS, with Python at the core. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Languages & Systems: Python (9yr), Bash, SQL, TypeScript, JavaScript, Linux (production 9yr), Concurrent Programming, Distributed Systems, Networking (TCP/IP), Git

Cloud & Delivery Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, Microservices, GitHub Actions CI/CD, RabbitMQ (event-driven async, retry logic, idempotency), Redis (caching), PostgreSQL, MySQL, Oracle, Automated Testing, Performance Engineering, Build Systems

AI & Observability: AI Model Deployment, LangGraph, LangChain, FastAPI, REST APIs, Observability & Metrics Pipelines, Model Evaluation, LLM Evaluation, Hallucination Detection, Dashboards, RAG Pipelines, ChromaDB, NLP, PyTorch, scikit-learn, PHI/HIPAA Compliance, Agile/Scrum

PROJECTS

AgentProbe: AI Model Delivery Infrastructure - Metrics Pipeline & CI Gate

Model Delivery · Observability · CI/CD · Deployment Gate

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions CI/CD · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production AI delivery infrastructure - metrics pipeline, model evaluation, and CI deployment gate that maintains complete visibility from model output to production. **Metrics pipeline:** 5-metric instrumentation engine (correctness, groundedness, hallucination rate, latency, token efficiency) tracking model behavior across every evaluation run; dashboards aggregating quality signals over time. **CI deployment gate:** GitHub Actions regression gate that blocks model quality degradation before deployment - tightening the feedback loop between model iteration and production evaluation. **Multi-model routing:** clean delivery abstraction for Anthropic, OpenAI, and Ollama - swap models without rewriting downstream systems. Zero regressions across 1,240+ test cases. Open-source, enterprise-deployable.

Leasel: Production AI System - Cloud Delivery & Distributed Pipeline

Production AI · Distributed · Cloud · CI/CD

Python · LangGraph · FastAPI · OCR · SQLite · GitHub Actions · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

End-to-end AI system built for production delivery reliability - cloud-deployable FastAPI backend, distributed 8-agent pipeline with fault-tolerant conditional routing, CI-integrated regression gate, and full audit logging across every execution path. Designed for the same production standards enterprise deployment infrastructure requires: idempotent operations, deterministic safety gates, versioned model evaluation, and complete observability of system behavior. 93.6% precision across 1,240 test cases; zero unsupported-claim rate.

AI on FHIR: Cloud-Native System Integration - Docker & AWS

Cloud-Native · Docker · AWS · REST APIs

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Cloud-native AI application containerized with Docker for scalable deployment - Python/Flask backend integrating AI inference with enterprise REST APIs; HIPAA-compliant RBAC; ~60% reduction in manual retrieval time. Demonstrates full cloud delivery lifecycle: containerized build, API integration layer, compliance-grade security, and production-ready deployment configuration.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Production AI Deployment

- Designed and deployed a production AI system (Python, LangGraph, FastAPI) for an external customer end-to-end - architecture, build, automated testing, CI/CD gate, and production deployment in weeks as sole technical owner; instrumented evaluation metrics and deployment validation to ensure model quality before release.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977); designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and systems design; translated complex computational outputs for non-technical stakeholders.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Cloud delivery infrastructure at production scale** - AWS S3/EC2/RDS/Glue production data pipelines; Docker microservices (50% scalability improvement, 30% deployment time reduction); GitHub Actions CI/CD with automated testing throughout; Python/Bash automation pipelines eliminating 80% of manual operational workload; 99.5% uptime across multi-tenant SaaS with zero security incidents across 7yr.
- **Event-driven distributed systems** - owned RabbitMQ async processing system for gross-rent updates across thousands of tenant records; designed retry logic, idempotency guarantees, and dead-letter handling that eliminated double-processing failures; Redis caching; 200+ SQL optimizations (30% performance gain); concurrent processing on Linux production systems throughout.
- **Observability and production reliability** - built internal debugging tools reducing triage time 40%; maintained bug count below 10 for 6+ consecutive months through proactive monitoring; owned production incident response for HUD compliance modules; 300+ peer code reviews; mentored 5+ engineers; leveraged Claude Code to accelerate engineering delivery velocity.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017